

Peiang (Ignatius) Miao

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RESEARCH INTERESTS

Multimodal agents, vision-language models, evidence-grounded reasoning, efficient and cost-aware inference, physical-world visual intelligence, digital dentistry, and clinical decision-support AI.

EDUCATION

The Hong Kong Polytechnic University

MSc, Metaverse Technology --- GPA 3.5/4.3

Department of Computing

Sep 2025 — Oct 2026

University College Dublin(joint program with Beijing University of Technology / BDIC)

B.Sc., Software Engineering---GPA 3.44/4.20, Upper Second Class Honours (parallel B.Eng.)

UCD China

Sep 2021 — Jun 2025

SELECTED PUBLICATIONS AND MANUSCRIPTS

Published

- **P. Miao**, S. Li, K. Cai, Y. Tian, R. Das, I. R. Nijhum, and S. Dev. “EmoSense: AFRN-RAG Framework for FER-Based Facial Emotion Analysis and Mental Health Support.” *33rd International Conference on Artificial Intelligence and Cognitive Science (AICS'25)*, Dublin. DOI: [10.5281/zenodo.19141746](https://doi.org/10.5281/zenodo.19141746).

Under Review

- X. Wei*, **P. Miao***, et al. “[SpaGVAE: A Variational Graph Autoencoder with a Structured Spatial Prior for Robust Spatial Domain Identification](#).” Selected by Cell Press editors for formal peer review at *Patterns* and *Cell Reports Methods*.
- X. Wei*, Y. Wang*, **P. Miao***, S. Mo, Y. Dong, R. Chen, H. Li, H. Lai, and S. Qian. “[Bio-CPD: A Distributional Divergence-based Framework for Single-cell Trajectory Analysis](#).” Under review at *iScience*.
- **P. Miao***, H. Wang*, D. Li, J. Li, and P. H. F. Ng. “Selective Vision-Language Inspection with Physics-Grounded Reasoning for Wafer Defect Recognition.” Under review at *IEEE Transactions on Cybernetics*.
- R. Das*, **P. Miao***, and S. Dev. “SS-VLM: A Spectral-Symbolic Vision-Language Framework for Auditable Facial Expression Recognition.” Under review at *IEEE Transactions on Artificial Intelligence*.

In Preparation

- **P. Miao***, H. Wang*, X. Wei*, and D. Li. “EviComp: Evidence-Calibrated Vision-Language Compositional Transfer for Composite Wafer Defect Recognition.” To be submitted to *IEEE Transactions on Industrial Informatics*.
- X. Wei*, **P. Miao***, et al. “COMPASS: Clinical Orchestration for Methodological Planning and Study Design” In preparation; corresponding authors: D. Li, H. Lai, J. Shi.
- **P. Miao***, X. Wei*, et al. “RA-REF: Reliability-Adaptive Relation-Evidence Fusion.” In preparation and to be submitted to *AAAI 2027*.

RESEARCH EXPERIENCE

Research Assistant / Co-First Author

The Hong Kong Polytechnic University

Mar 2026 — Present

Hong Kong SAR

- Supervised by [Dr. Dun Li](#), [Dr. Huan wang](#) and [Dr. Peter HF Ng](#)
- Developed **EviComp**, an evidence-calibrated VLM framework for composite wafer-map recognition that predicts component sets and component-specific heatmaps rather than a single dominant defect label. Designed a component-query evidence decoder with mask supervision, semantic alignment, and calibrated target-domain head adaptation to separate synthetic component learning from real-domain benchmark transfer.
- Constructed **CrystalComp**, a counterfactual compositional wafer benchmark with component labels, spatial evidence masks, and held-out component combinations for controlled compositional generalization.
- Built **PGC-Agent**, a progressive wafer inspection framework combining uncertainty-gated routing, LoRA-tuned VLM verification, and a physics-grounded reasoning agent for hard mixed-defect cases. Implemented physical evidence tools including 2D-FFT spectra, Radon transforms, radial profiles, KDE density maps, polar views, and structured evidence reports for auditable industrial reasoning.

Co-First Author / Computational Contributor

Shanghai Ninth People's Hospital, Shanghai Jiao Tong University, and PolyU

May 2026 — Present

Remote, Hong Kong SAR

- Supervised by [Prof. Hongchang Lai](#) and [Prof. Junyu Shi](#).
- Co-developed **Bio-CPD**, detects cliff points (tipping points / critical transitions) along continuous single-cell trajectories such as pseudotime, developmental time, or aging. It scans a sliding window across the trajectory, computes a combined Cliff Point Index (CPI) from Bhattacharyya distance and Wasserstein distance with entropy-based adaptive weighting, identifies peaks, and validates each candidate via logistic-regression AUROC against null distributions — all in a single function call.
- Co-first-authored **SpaGVAE**, supporting method development, benchmarking, manuscript preparation, and public code release for a Cell Press manuscript selected for formal peer review. Implemented entropy-weighted Bhattacharyya/Wasserstein fusion, AUROC-based transition validation, ablation studies, and cross-dataset benchmarking against established change-point methods.
- Developing **CADENCE**, a clinical-dental LLM agent for transforming implant-related research ideas into structured protocols with evidence retrieval, causal-boundary checks, feasibility review, and expert-facing critique.

Research Assistant

University College Dublin

Dec 2024 — Jun 2026

Remote, Beijing, China

- Supervised by [Riju Das](#) and [Associate Prof. Soumyabrata Dev](#).
- Developed **EmoSense**, an end-to-end pipeline integrating real-time facial-expression recognition at 30 fps with retrieval-augmented temporal emotion summarization and mental-health-oriented report generation. Designed an Adaptive Feature Recalibration Network (AFRN) with dual-pathway attention and progressive feature fusion, achieving 95.39% accuracy on CK+.
- Co-first-authored **SS-VLM**, a spectral-symbolic VLM for auditable facial expression recognition using Fourier-domain visual cues, learned Action Unit priors, retrieval-grounded symbolic reasoning, and constrained report generation. SS-VLM reframes facial expression recognition from a black-box label prediction task into an auditable affective reasoning pipeline.

SELECTED TECHNICAL PROJECTS

Co-Founder

Beijing Shawarma Tech (SAGA AI)

Jun 2025 — Present

Beijing, China

- Built FastAPI backend services for retrieval-augmented generation, AI-authorship detection, streaming LLM responses, and interactive editing assistance.
- Integrated third-party detectors with LLM-driven feature analysis to generate document-level and sentence-level feedback for writing assessment workflows.
- Implemented vector retrieval pipelines with Milvus for embedding storage, semantic search, and evidence-backed feedback generation.

Software Engineer Intern

Tianjin Shengtong Science and Technology Development Co. Ltd.

Jun 2024 — Aug 2024

Tianjin, China

- Developed a B/S three-tier management system for 200+ enterprise users, automating 5+ back-office processes and reducing manual reporting time by 40%.

Database Developer Intern

Haibin People's Hospital of Tianjin Binhai New Area

Jun 2023 — Aug 2023

Tianjin, China

- Optimized hospital database workflows for 50,000+ patient records, reducing redundancy by 25% and improving query speed by 18%.

SKILLS

- **Languages:** English (IELTS 6.5; BSc and MSc programmes conducted entirely in English), Mandarin (native proficiency).
- **Programming:** Python, MATLAB, JavaScript, SQL, Java.
- **Machine Learning:** PyTorch, TensorFlow, vision-language models, retrieval-augmented generation, multimodal learning, uncertainty-aware routing.
- **Computer Vision:** defect recognition, facial expression recognition, spectral analysis, segmentation, heatmap/evidence supervision, 3D/digital-twin research planning.
- **Tools:** FastAPI, Milvus, Git, LaTeX, Typst.

AWARDS AND CERTIFICATES

- Honorable Mention, Mathematical Contest in Modeling / Interdisciplinary Contest in Modeling, 2023.
- Best Final Year Project (ranked 1/96), BSc Software Engineering, University College Dublin, 2025.
- Deep Learning Specialization, Coursera. [Certificate](#).
- Natural Language Processing Specialization, Coursera. [Certificate](#).